Chapter: 6

I. Hash Function

A hash function H acknowledges a variable-length square of information as info and produces a fixed-size hash esteem.. Hash function provide data integrity. A little changes in a result's bit will create a high changes in hash code.

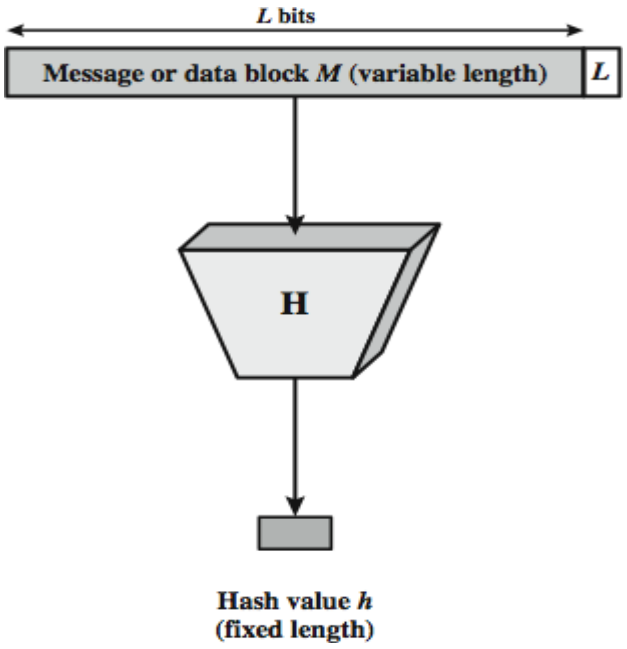


Fig 6.1 Hash Function [1]

- A cryptographic hash capacity can characterize as a calculation for which it is computationally infeasible to discover -
  - The one-way property
  - Two information protests that guide to a similar hash result (the collision-free property).

II. Applications of Cryptographic Hash Functions

- Message authentication
- Digital Signature
- One-way password file

III. Message Authentication

- MA is an instrument or administration used to confirm the trustworthiness of a message.
- MA guarantees that information got are actually as sent (i.e., contain no change, inclusion, cancellation, or replay).
- When a hash work is utilized to give message verification, the hash work esteem is regularly alluded to as a message digest.

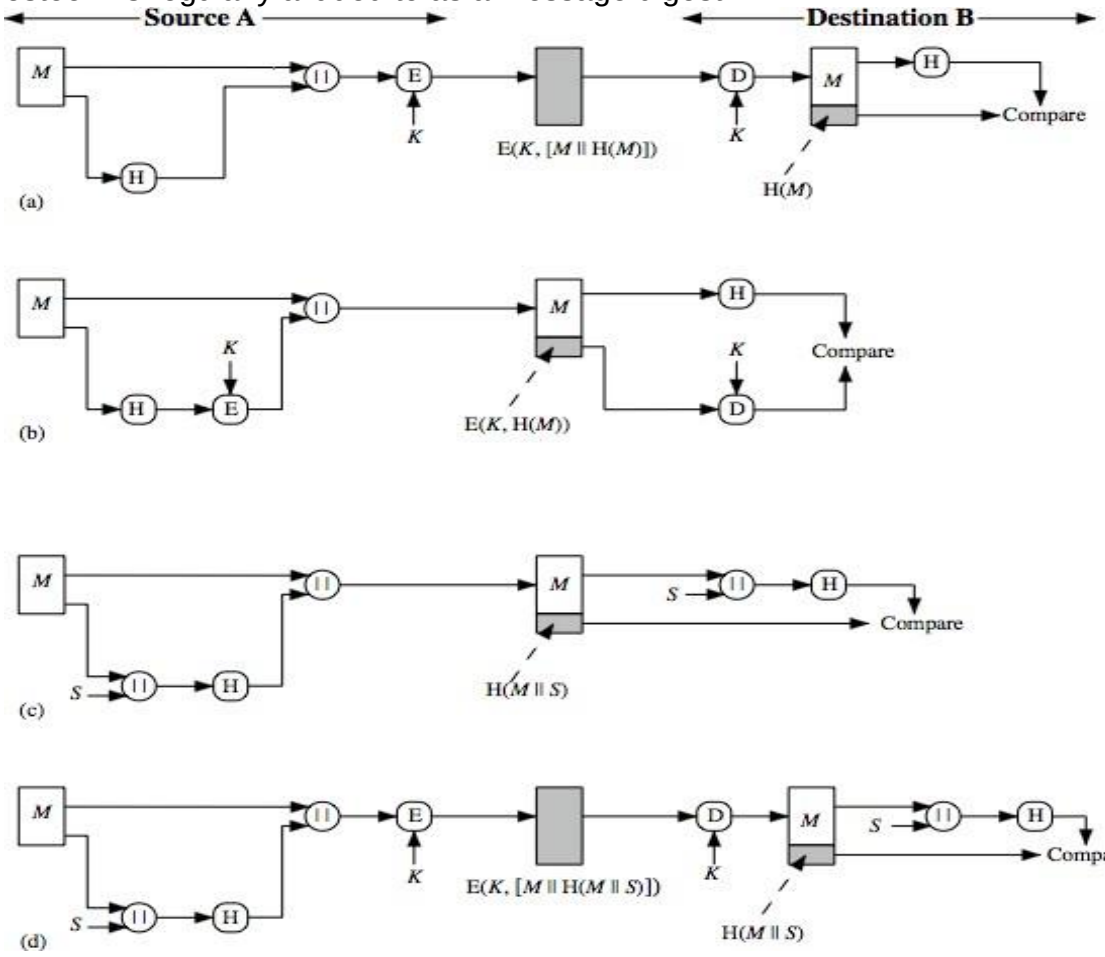


Fig.6.2 Message Authentication Methods

IV. Requirements for hash functions

- Can be applied to any sized message M.
- Produces fixed-length output h.
- It is easy to compute h=H(M) for any message M.
- Given hash value h is infeasible to find y such that (H(y) = h) is One-way property

- For given block x, it is computational infeasible to find y ≠ x with H(y) = H(x) is Weak collision resistance
- It is computationally infeasible to find messages m1 and m2 with H(m1) = H(m2) is Strong collision resistance

Simple Hash Function

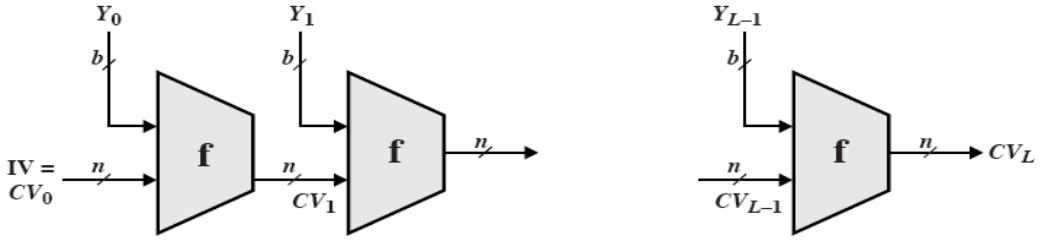
- The information like message, document is seen as a succession of n-bit blocks.
- The information is prepared each square in turn in an iterative design to create a n-digit hash work.
- One of the least difficult hash capacities is the step by step restrictive OR (XOR) of each square.

$$C_i = b_{i1} \oplus b_{i2} \oplus \ldots \oplus b_{im}$$

Where,

C_i = ith bit of the hash code 1 ≤ i ≤ n
m = number of n-bit blocks in the input
b_{ij} = ith bit in jth block

General Structure of Secure Hash Code



IV = Initial value
CV = chaining variable
Y_i = ith input block
f = compression algorithm
L = number of input blocks
n = length of hash code
b = length of input block

Fig.6.3 General Structure of Secure Hash Code

v. MD5: Message Digest Version 5

- MD5 Specified as Internet standard RFC1321

- **MD5 Overview**
  - Pad message so its length is 448 mod 512

- Append a 64-bit original length value to message
- Initialise 4-word (128-bit) MD buffer (P,Q,R,S)
- Process message in 16-word (512-bit) blocks:
- Using 4 rounds of 16 bit operations on message block & buffer
- Add output to buffer input to form new buffer value
- Output hash value is the final buffer value

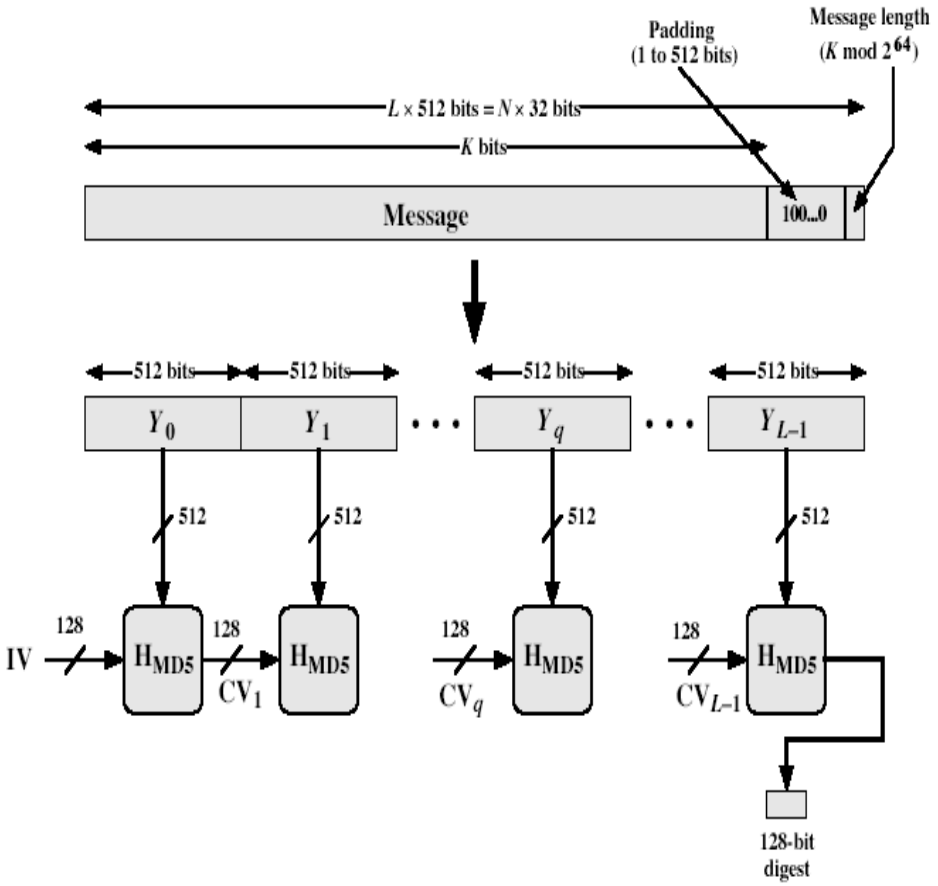


Fig.6.4 Message Digest Version 5

Padding Twist

- Step1: Given unique message M, add cushioning pieces "10*" with the end goal that subsequent length is 64 pieces not exactly a numerous of 512 pieces.
- Step2: Append (unique length in bits mod 264), spoke to in 64 pieces to the cushioned message

- Final message is hacked 512 pieces a square

MD5 Process

- As many stages as the number of 512-bit blocks in the final padded message
- Digest: 4 32-bit words: MD=P|Q|R|S
- Every message block contains 16 32-bit words: m0|m1|m2…|m15
  - Digest MD0 initialized (IV) to: used little –endian methods P=01234567,Q=89abcdef,R=fedcba98, S=76543210
  - Every stage consists of 4 passes over the message block, each modifying MD

Each block 4 rounds, each round 16 steps

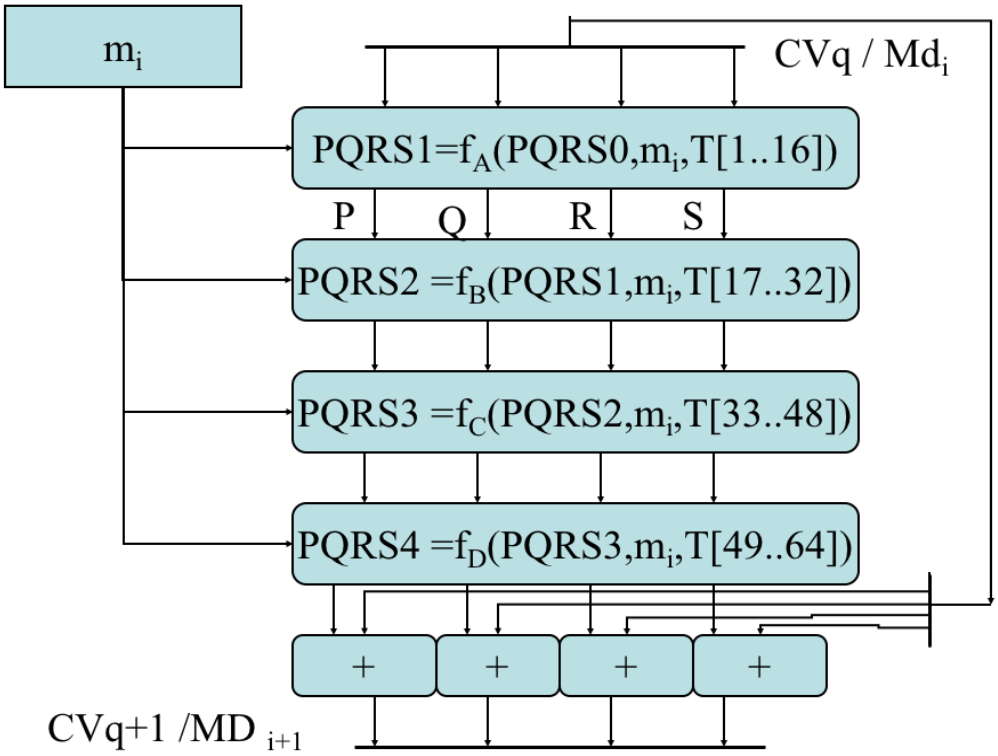


Fig.6.5 MD5 Process

VI. Secure Hash Algorithm –SHA

- Created by NIST, indicated in the Secure Hash Standard (SHS, FIPS Pub 180), 1993
- SHA is indicated as the hash calculation in the Digital Signature Standard (DSS), NIST
- SHA initially planned by NIST and NSA in 1993 was changed in 1995 as SHA-1



- It depends on plan of MD4 with key contrasts
- It Produces 160-piece hash esteems
- NIST gave modification FIPS 180-2 of every 2002 and adds 3 extra forms of SHA i.e SHA-256, SHA-384, SHA-512
- Structure and detail is like SHA-1 subsequently examination ought to be comparable
- But security levels are somewhat higher

Table 6.1 SHA

	SHA - 1	SHA - 224	SHA - 256	SHA - 384	SHA - 512
Message Digest Size	160	224	256	384	512
Message Size	$< 2^{64}$	$< 2^{64}$	$< 2^{64}$	$< 2^{128}$	$< 2^{128}$
Block Size	512	512	512	1024	1024
Word Size	32	32	32	64	64
Number of Steps	80	64	64	80	80

Secure Hash Algorithm-1 (SHA-1)

Basic Steps-

Step1: Padding

Step2: Appending length as 64 bit unsigned

Step3: Initialize MD buffer 5 32-bit words

Store in big endian format, most significant bit in low address P|Q|R|S|T

P = 67452301 Q = efcdab89

R = 98badcfe S = 10325476 T = c3d2e1f0

Step 4: The 80-step processing of 512-bit blocks – 4 rounds, 20 steps each.

Each step *t* (0 <= *t* <= 79):

Input:

- W_t – a 32-bit word from the message
- K_t – a constant.
- PQRST: current MD.

Output:

- PQRST : new MD.

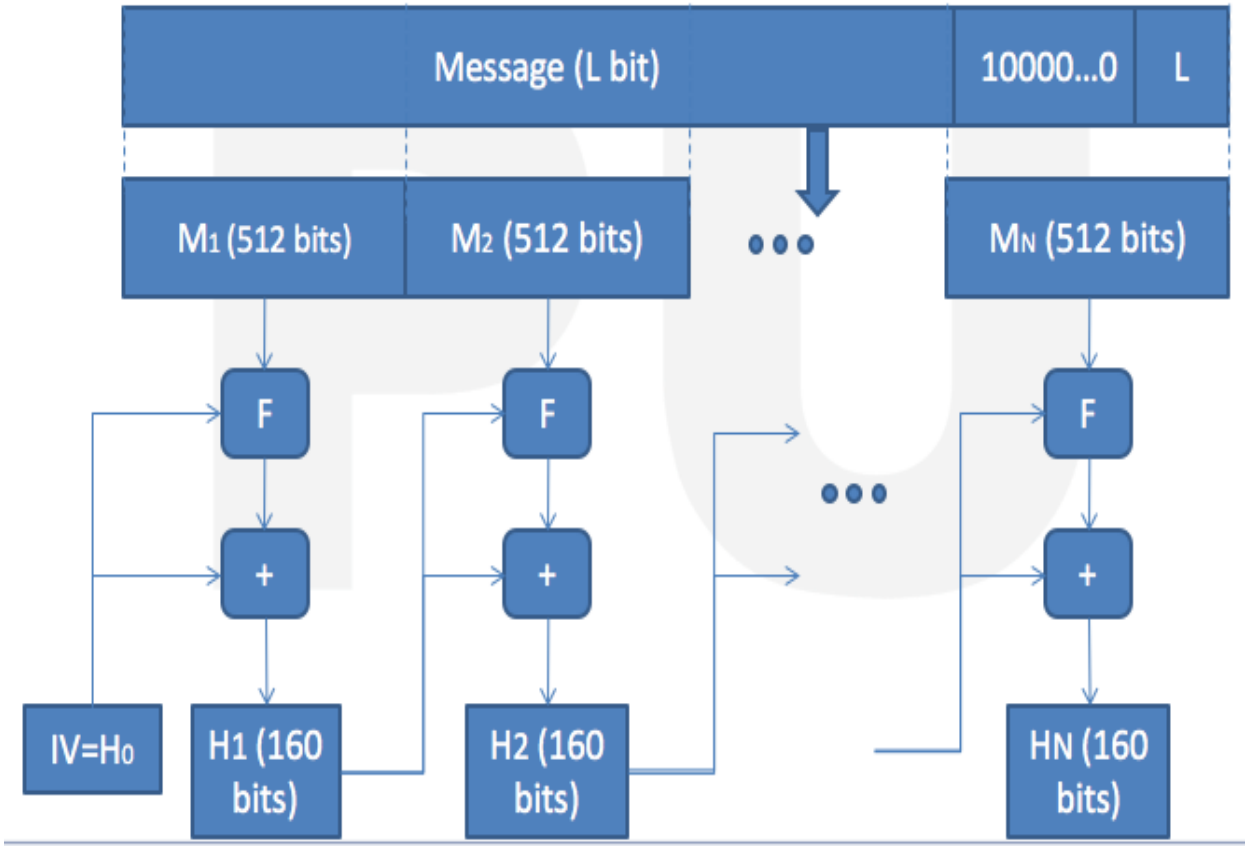


Fig.6.6 SHA-1

VII. Secure Hash Algorithm-512 (SHA-512)

- The algorithm takes as input a message with a maximum length of less than 2128 bits and produces as output a 512-bit message digest.
- The input is processed in 1024-bit blocks.

Step -1 Append Padding Bits

- The message is cushioned so its length is harmonious to 896 modulo 1024 [length ≡ 896(mod 1024)] .



- Padding is constantly added, regardless of whether the message is now of the ideal length.
- Thus, the quantity of cushioning pieces is in the scope of 1 to 1024.
- The cushioning comprises of a solitary 1 bit followed by the important number of 0 pieces.

Step -2 Append Length

Step -3 Initialize hash buffer

- The result of the initial two stages creates a message that is a whole number numerous of 1024 pieces long.
- The extended message is spoken to as the arrangement of 1024-digit blocks M1 , M2, … , MN, with the goal that the all out length of extended message is N X 1024 pieces.
- A 512-digit cushion is utilized to hold transitional and eventual outcomes of the hash work. The cushion can be spoken to as eight 64-bit enrolls (a, b, c, d, e, f, g, h).

Step -4 Process message in 1024-bit (128-word) blocks

- The core of the calculation is a module that comprises of 80 adjusts; this module is named F

Step – 5 Output

- After all N 1024-bit blocks have been handled, the yield from the Nth stage is the 512-cycle message digest

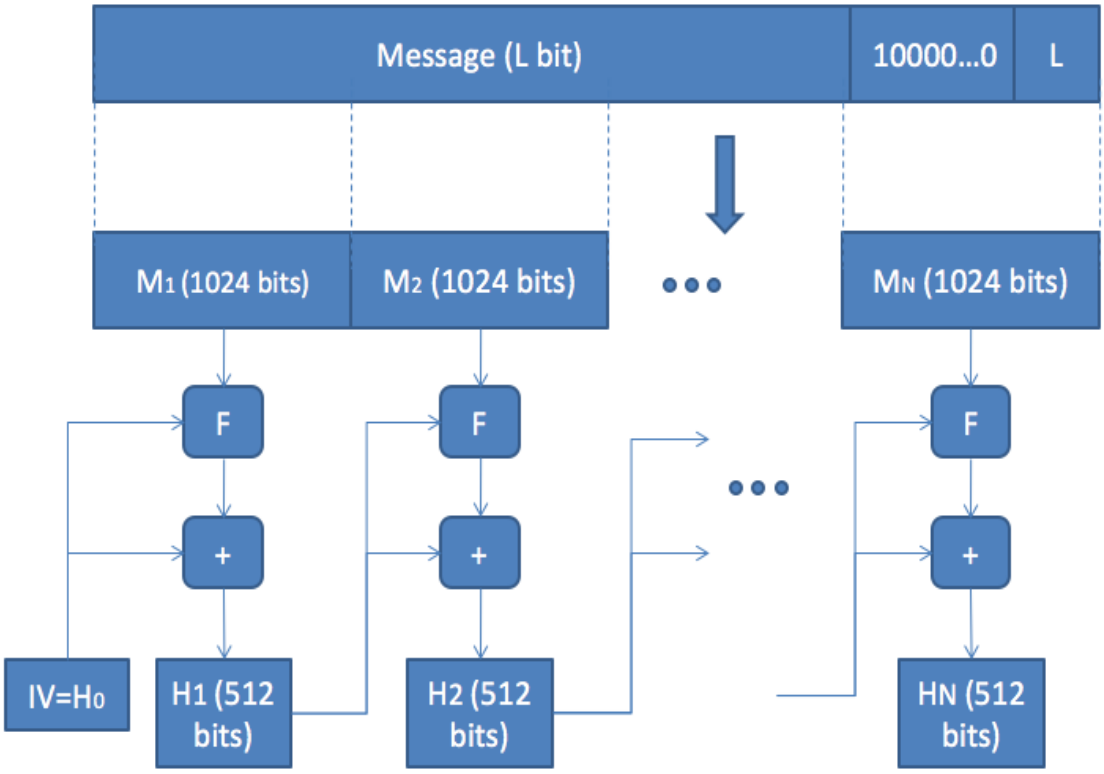


Fig.6.7 SHA-512

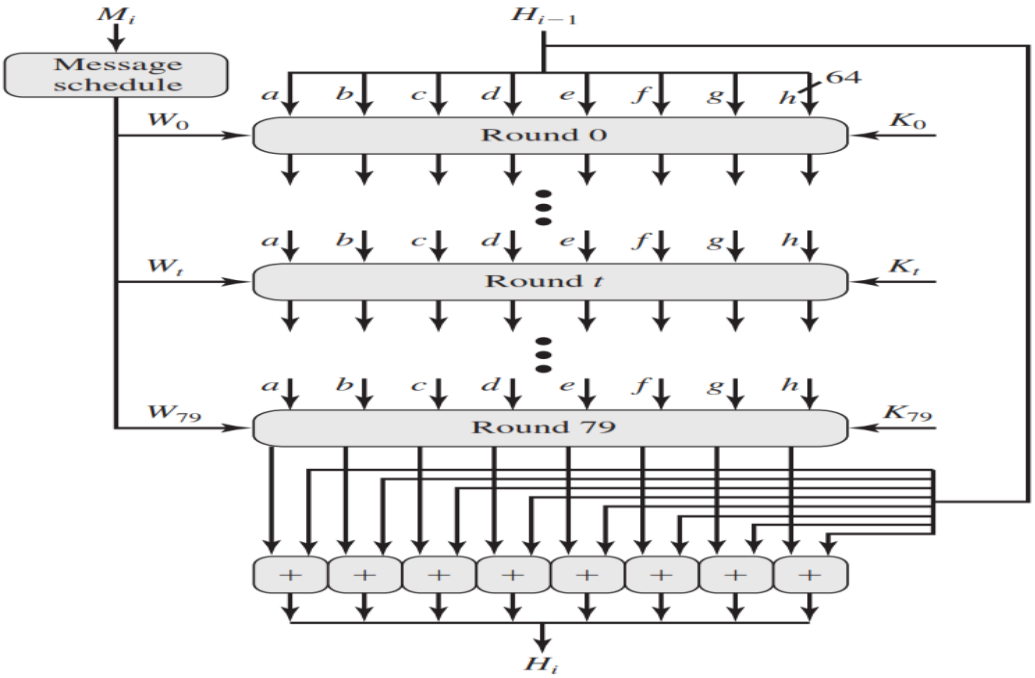


Fig.6.8 SHA-512 Processing of a Single 1024-Bit Block

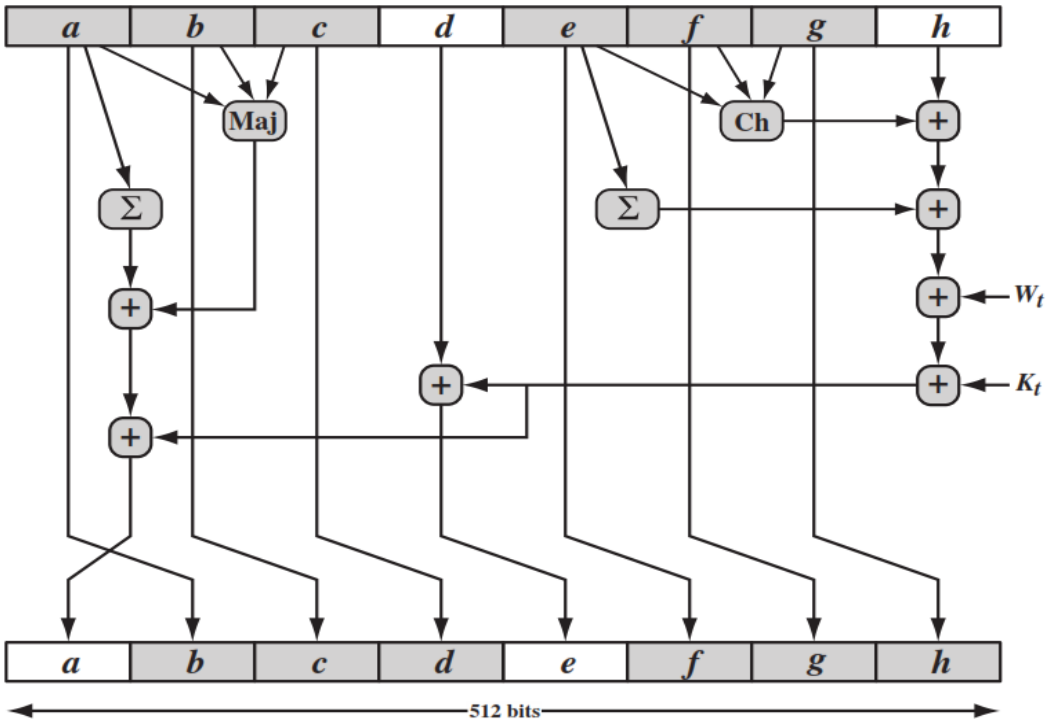


Fig.6.9 SHA-512 Round Function

SHA-512 Round Function Elements

- Maj(a,b,c) = (a AND b) XOR (a AND c) XOR (b AND c) Majority of arguments are true
- ∑(a) = ROTR(a,28) XOR ROTR(a,34) XOR ROTR(a,39)
- ∑(e) = ROTR(e,14) XOR ROTR(e,18) XOR ROTR(e,41)
- + = addition modulo 2^{64}
- K_t = a 64-bit additive constant
- W_t = a 64-bit word derived from the current 512-bit input block.

SHA-1 v/s MD5

- Brute power assault is more enthusiastically (160 versus 128 pieces



for MD5)

- Not defenseless against any known cryptanalytic assaults (contrasted with MD4/5)
- A little more slow than MD5 (80 versus 64 stages)
- Both function admirably on a 32-digit design
- Both planned as straightforward and minimal for execution

Message Authentication

- Message validation is a technique to check that got messages come from the veritable source and have not been modified.
- Message verification may likewise confirm sequencing and idealness.
- Message verification is a system or administration used to check the trustworthiness of a message.
- Message verification guarantees that information got are actually as sent

Message Authentication Requirements

- Disclosure: Release of message substance
- Traffic analysis: Discovery of the example of traffic between parties
- Masquerade: Insertion of messages into the organization from a fake source
- Content modification: Changes to the substance of a message
- Sequence modification: Any alteration to an arrangement of messages between parties
- Timing modification
- Source repudiation: Denial of transmission of message by source
- Destination repudiation: Denial of receipt of message by objective

Authentication Functions

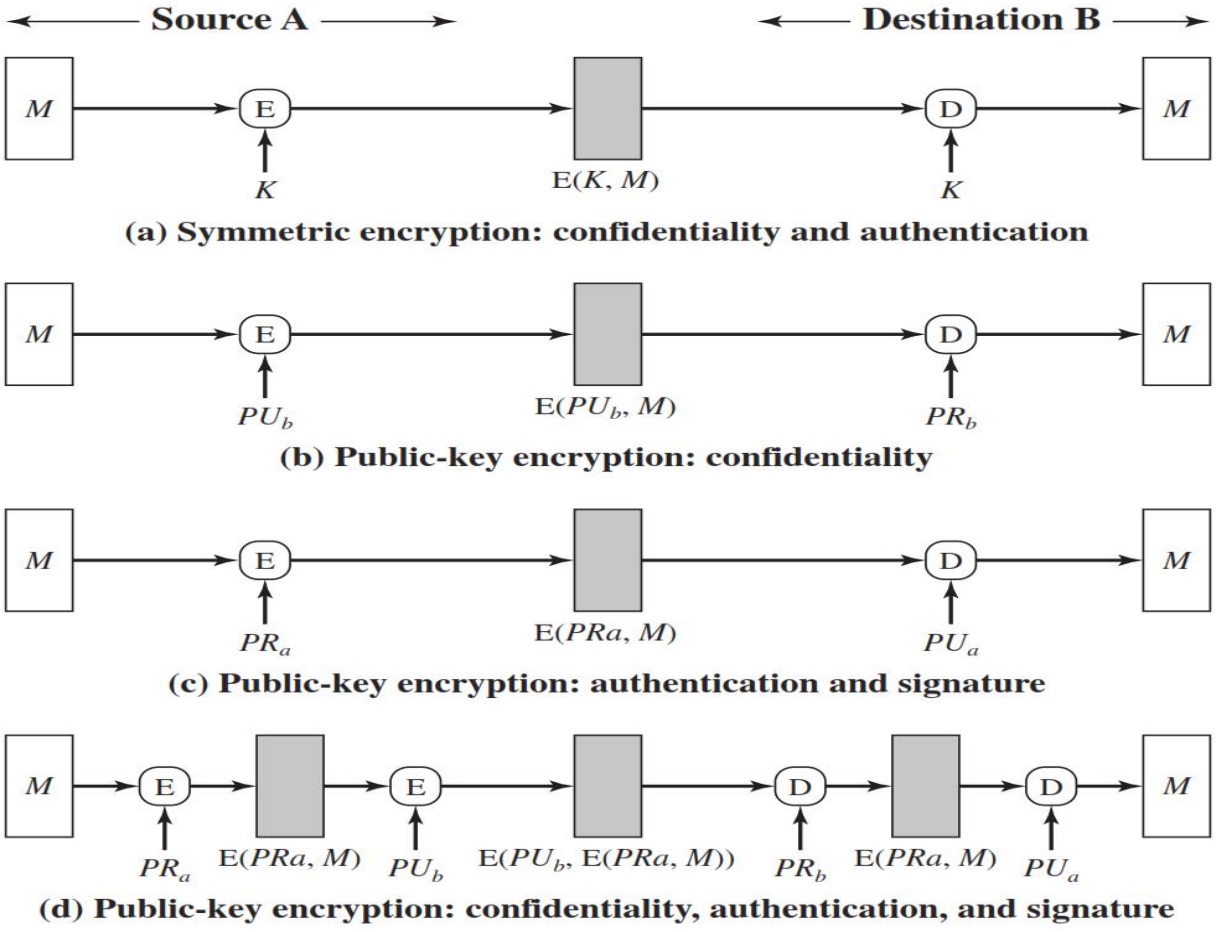


Fig.6.10 Authentication Functions

Message Authentication Code

- MAC incorporates the use of a riddle key to make a little fixed-size square of data, known as a cryptographic checksum or MAC
- MAC is annexed to the message. This procedure expects that two imparting parties, state An and B, share a typical mystery key K.
- When A has a message to ship off B, it ascertains the MAC as a component of the message and the key

$$MAC = C (K , M)$$

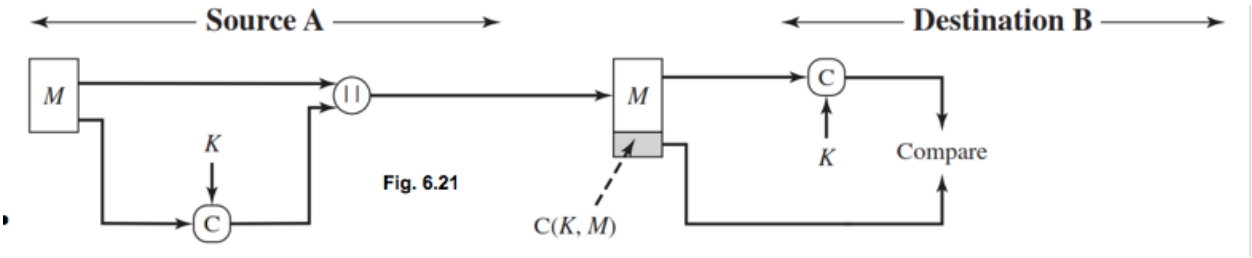


Fig. 6.21

Fig.6.11 Message Authentication

VIII. MAC Based on Hash Functions – HMAC

- Cryptographic hash capacities, for example, MD5 and SHA by and large execute quicker in programming than symmetric square codes, for example, DES.
- Library code for cryptographic hash capacities is broadly accessible.

Design objectives for HMAC

- To use, without adjustments, accessible hash capacities.
- To take into consideration simple supplant capacity of the inserted hash work on the off chance that quicker or safer hash capacities are found or required.
- To protect the first presentation of the hash work without acquiring a critical corruption.
- To utilize and deal with keys in a straightforward manner.
- To have a surely known cryptographic examination of the quality of the confirmation system dependent on sensible presumptions about the installed hash work.

HMAC Structure

- Append zeros to the left end of K to create a b-bit string K+
- XOR K+ with ipad to produce the b-bit block Si.
- Append M to Si.
- Apply H to the stream generated in step 3.

- XOR K+ with opad to produce the b-bit block S0.
- Append the hash result from step 4 to S0.
- Apply H to the stream generated in step 6 and output the result.

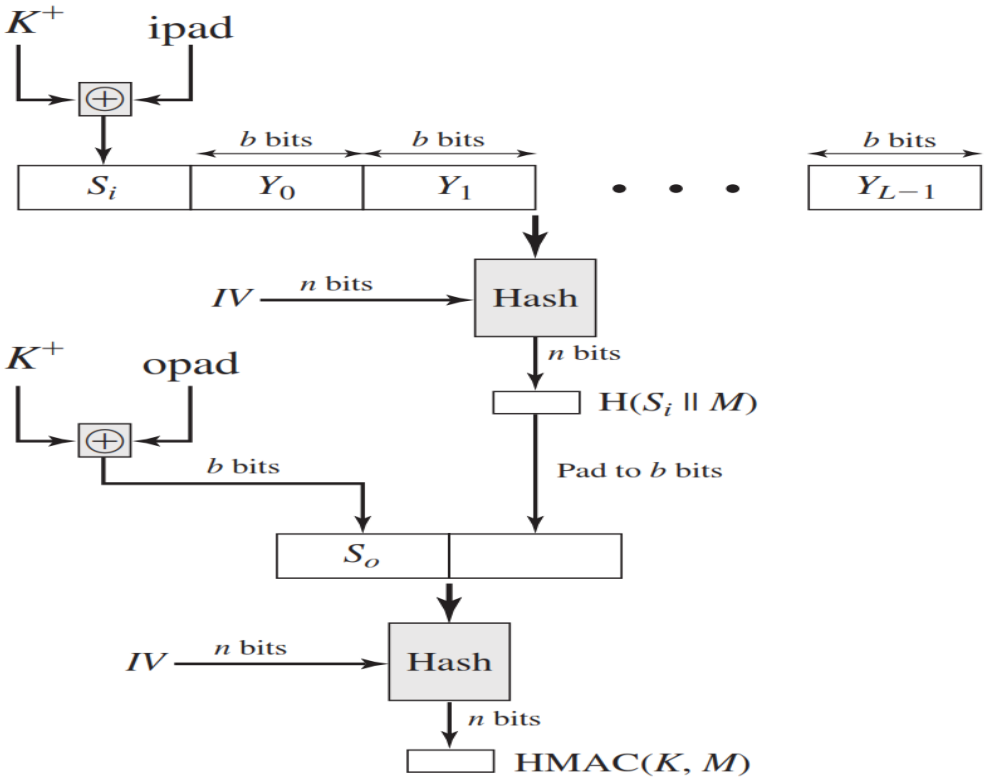


Fig.6.12 HMAC

- H = embedded hash function (e.g., MD5, SHA-1, RIPEMD-160)
- IV = initial value input to hash function
- M = message input to HMAC
- Yi = i th block of M
- L = number of blocks in M
- n = length of hash code produced by embedded hash function
- K + = K padded with zeros on the left so that the result is b bits in

length

- ipad = 00110110 (36 in hexadecimal) repeated b/8 times
- opad = 01011100 (5C in hexadecimal) repeated b/8 times

IX. MAC based on Block Ciphers

- The Data Authentication Algorithm (DAA), in light of DES, has been one of the most broadly utilized MACs for various years.
- The calculation can be characterized as utilizing the code block anchoring (CBC) method of activity of DES with an IV of zero

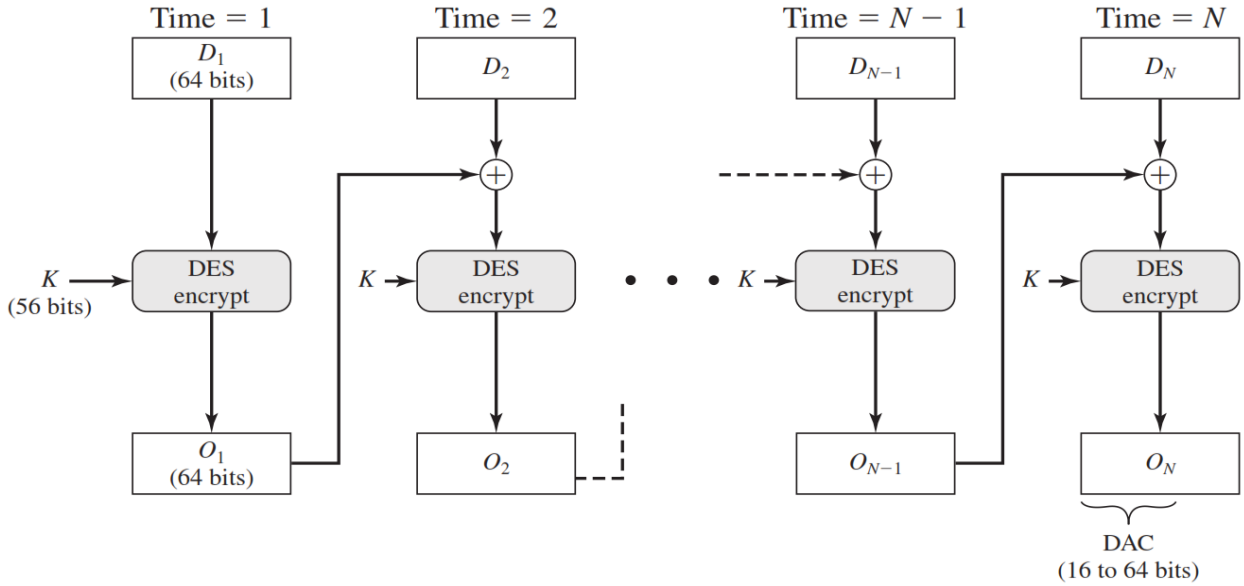


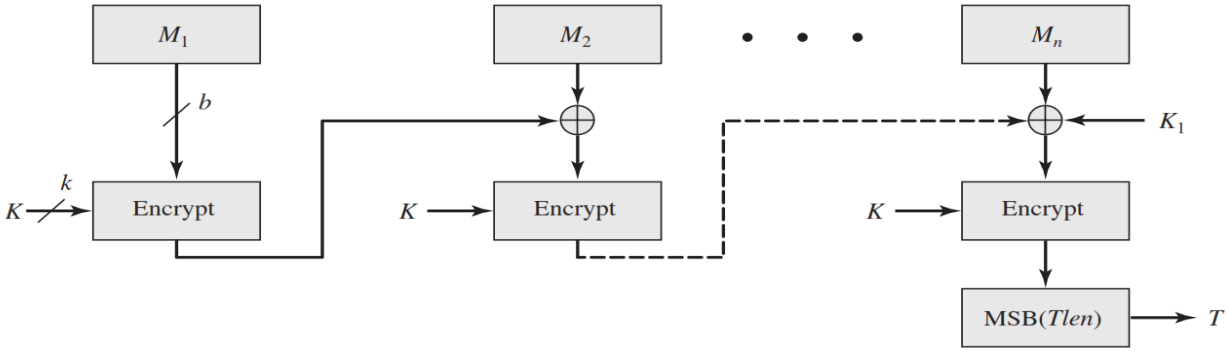
Fig.6.13 DAA

- The information to be validated are gathered into coterminous 64-cycle blocks:
- D1, D2, … . Dn. In the event that fundamental, the last square is cushioned on the privilege with zeroes to frame an entire 64-digit block. Utilizing the DES encryption calculation E and a mystery key K, an information verification code (DAC) is determined as follows

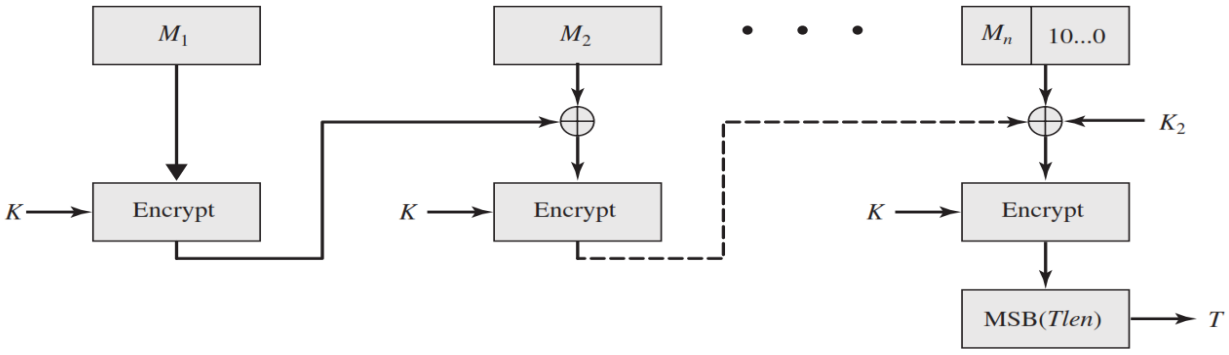
$$
\begin{aligned}
O_1 &= \mathrm{E}(K, D) \\
O_2 &= \mathrm{E}(K, [D_2 \oplus O_1]) \\
O_3 &= \mathrm{E}(K, [D_3 \oplus O_2]) \\
&\cdot \\
&\cdot \\
&\cdot \\
O_N &= \mathrm{E}(K, [D_N \oplus O_{N-1}])
\end{aligned}
$$

Cipher-Based Message Authentication Code (CMAC)

- Cipher-based Message Authentication Code (CMAC) method of activity for use with AES and triple DES.
- The calculation utilizes a k-bit encryption key K and a b-bit steady, K1.
- For AES, the key size k is 128, 192, or 256 pieces; for triple DES, the key size is 112 or 168 pieces.



(a) Message length is integer multiple of block size



(b) Message length is not integer multiple of block size

Fig.6.14 CMAC